

MSTAR PRACTICE WORKSHEET

Lesson 4-4: Find and Compare with Percentages

Grade 6 Mathematics · Standard 6.AT.4

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 **SELECTED RESPONSE — CORRECT: B**

What is 20% of 150?

- ☐ A 20
- ☒ B 30
- ☐ C 35
- ☐ D 75

20% = 0.20. Multiply: $0.20 \times 150 = 30$.

Question 2 **SELECTED RESPONSE — CORRECT: B**

What is 50% of 84?

- ☐ A 34
- ☒ B 42
- ☐ C 50
- ☐ D 168

50% = 0.50. Multiply: $0.50 \times 84 = 42$. (Or simply divide by 2: $84 \div 2 = 42$.)

Question 3 SELECTED RESPONSE — CORRECT: B

What is 10% of 320?

- ☐ A 3.2
- ☒ B 32
- ☐ C 64
- ☐ D 320

10% = 0.10. Multiply: $0.10 \times 320 = 32$. (Or move the decimal one place left.)

Question 4 SELECTED RESPONSE — CORRECT: B

What percent of 50 is 15?

- ☐ A 15%
- ☒ B 30%
- ☐ C 35%
- ☐ D 50%

$15/50 \rightarrow 50 \times 2 = 100$, so $15 \times 2 = 30$. That makes $15/50 = 30/100 = 30\%$.

Question 5 SELECTED RESPONSE — CORRECT: C

An arcade prize costs 240 tickets. If you have a coupon for 15% off, how many tickets do you save?

- ☐ A 15 tickets
- ☐ B 24 tickets
- ☒ C 36 tickets
- ☐ D 204 tickets

15% of 240 = $0.15 \times 240 = 36$ tickets saved.

Question 6

SELECTED RESPONSE — CORRECT: C

A student scored 72 out of 90 on a test. What percent did they score?

- ☐ A 72%
- ☐ B 75%
- ☒ C 80%
- ☐ D 90%

72/90 simplifies to 8/10, and $10 \times 10 = 100$, so $8 \times 10 = 80$. That makes $72/90 = 80/100 = 80\%$.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7

TWO-PART QUESTION

Part A — correct answer: C

At West Middle School, 45% of the 360 sixth-grade students participate in after-school sports. How many sixth-grade students participate in sports?

- ☐ A 144 students
- ☐ B 160 students
- ☒ C 162 students
- ☐ D 180 students

$45\% \text{ of } 360 = 0.45 \times 360 = 162 \text{ students.}$

- **A:** That is 40% of 360. Multiply by the actual rate: 0.45×360 .
- **B:** That is an estimate rather than the exact product. 0.45×360 comes to 162.
- **D:** That is 50% of 360. The rate is 45%, a little less than half.

Part B — correct answer: A

Which calculation method accurately justifies this solution in Part A?

- ☒ A Multiply the decimal representation of the percent by the whole: $0.45 \times 360 = 162$.
- ☐ B Subtract 45 from 360: $360 - 45 = 315$.
- ☐ C Divide 360 by 45: $360 \div 45 = 8$.
- ☐ D Multiply 360 by 4.5: $360 \times 4.5 = 1,620$.

Finding a percent of a number requires multiplying the rate per hundred (0.45) by the total population (360).

Question 8

WRITTEN RESPONSE · 2 POINTS

Type II Reasoning: Compare Percent Discounts

Store A offers a \$120 skateboard at 30% off. Store B offers the same \$120 skateboard with a \$40 instant manufacturer coupon.

Which store offers the lower final price? Justify your decision by showing the savings and final price at both stores.

Model answer: Store B offers the lower final price (\$80 vs \$84). Store A discount is 30% of \$120 = \$36 (final price \$84). Store B discount is \$40 (final price \$80). Store B saves \$4 more.

2 points	Student calculates Store A discount as \$36 (0.30×120) with final price \$84, calculates Store B final price as \$80 ($\$120 - \40), and concludes Store B is cheaper by \$4.
1 point	Student correctly calculates discounts for both stores but makes an arithmetic error on final price or fails to state which store is cheaper.
0 points	Student chooses a store without supporting mathematical calculations.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9

WRITTEN RESPONSE · 2 POINTS

A class has 32 students. 25% are in the math club, 50% are in the reading club, and 12.5% are in both. How many students are in each group? Can the total of all three groups be more than 32? Explain why or why not.

Model answer: Taking each percent of the same 32 students gives $25\% = 32 \div 4 = 8$ in math club, $50\% = 32 \div 2 = 16$ in reading club, and $12.5\% = 32 \div 8 = 4$ in both. Adding $8 + 16 + 4 = 28$ is misleading, because the 4 students in both are already inside the 8 and the 16 — counting them a third time overlaps, and the number of different students in at least one club is $8 + 16 - 4 = 20$.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.